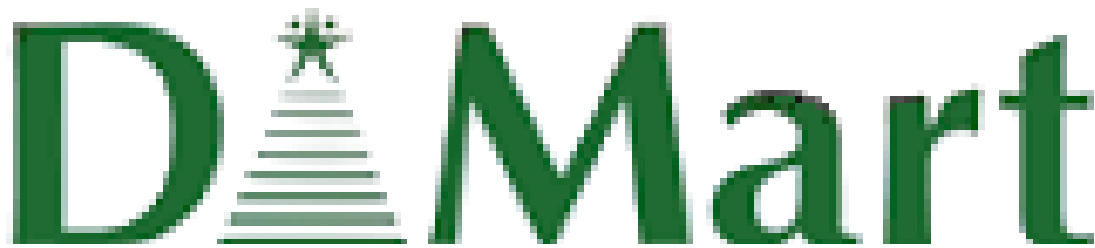


VLOOKUP, HLOOKUP & XLOOKUP

- Why Use Lookup Functions
- VLOOKUP vs HLOOKUP vs XLOOKUP
- Practical Use Cases & Formulas



Why Use Lookup Functions?

Efficiently retrieve data from large datasets without manual searching

VLOOKUP (Vertical Lookup)

Searches for a value in the first column and returns a value from a specified column.

Syntax: =VLOOKUP(lookup_value, table_array, col_index, [range_lookup])

Example: =VLOOKUP("ORD100000", A2:L15346, 4, FALSE) returns the price of Product A

HLOOKUP (Horizontal Lookup)

Searches for a value in the first row and returns a value from a specified row.

Syntax: =HLOOKUP(lookup_value, table_array, row_index, [range_lookup])

Example: =HLOOKUP("D1", A1:L1, 4, FALSE) returns D1 sales from row 4

XLOOKUP (Modern Replacement)

Searches any column/row and returns a match from any position. Replaces both VLOOKUP & HLOOKUP.

Syntax: =XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found])

Example: =XLOOKUP("ORD100000", B2:L15346, 0)

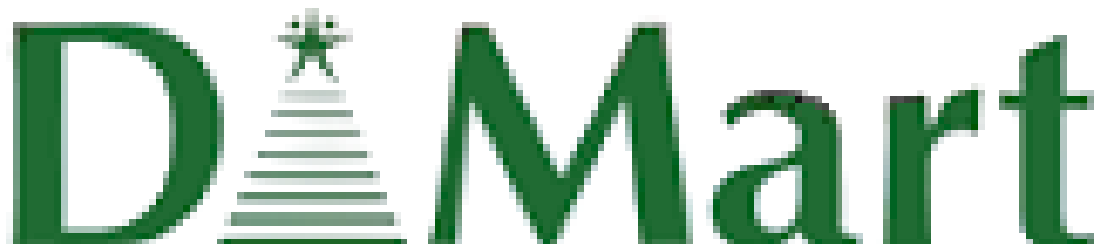
When to Use Each Function

VLOOKUP = Vertical tables | HLOOKUP = Horizontal tables | XLOOKUP = Any direction

VLOOKUP: lookup column must be leftmost (limited)

HLOOKUP: lookup row must be topmost (limited)

XLOOKUP: no restrictions, supports exact/approximate/wildcard matches



Comparison & Best Practices

VLOOKUP Limitations

Lookup column must be the leftmost column in the range.
Cannot search to the left; only returns values from columns to the right.

HLOOKUP Limitations

Lookup row must be the topmost row in the range.
Best suited for small horizontal tables; rarely used in large datasets.

XLOOKUP Advantages

Searches in any direction. Built-in error handling with [if_not_found] parameter.
Supports exact, approximate, and wildcard match modes natively.

Performance Considerations

XLOOKUP is binary-search capable for sorted data, making it faster on large datasets.
VLOOKUP with FALSE (exact match) scans linearly and slows on 100K+ rows.

Recommendation Summary

New projects => Use XLOOKUP (modern, flexible, error-safe)
Legacy files => VLOOKUP still works; migrate when convenient
Horizontal tables => HLOOKUP or XLOOKUP with row arrays

Key Takeaway

XLOOKUP is the recommended default for all new work in Excel 365 and Google Sheets.
It combines the best of VLOOKUP and HLOOKUP with fewer limitations.